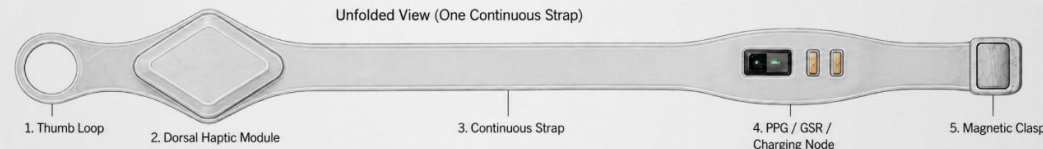




Wearable Wristband Based on Vision-Language Model Environmental Interpretation

Assisting Independent Mobility
for Low-Vision Users

Theme Practice



USER RESEARCH

Heideng

A professional stand-up comedian, he has a lot of fans in China.

Basic Profile

Occupation	Stand-up comedian and rare-disease advocate
Vision	Stargardt disease, with visual acuity of approximately 0.02. Severe central vision loss; mainly perceives blurred outlines
Digital literacy	Photographs distant text and enlarges it on his phone. Uses OCR and text-to-speech to access visual information.
Mobility	Memorises metro routes, stations and transfers in advance. Moves closer to signs or uses his phone to recognise text.
Personality	Independent, candid, humorous and observant; dislikes unnecessary assistance

GOALS

This user's goals were developed from publicly available interviews and media reports.

1. Maintain independence and control over everyday mobility.
2. Access the most important environmental information quickly.
3. Reduce repeated phone-based interactions.
4. Move efficiently when travelling between performances.
5. Receive assistance without being controlled, labelled or medicalised by the product.
6. Preserve personal style and a dignified social identity.

DEMANDS DIG

Functional Needs

- Clear directional guidance.
- Timely alerts for obstacles, steps, turns and route deviation.
- Recognition of seat numbers, door signs, platforms and essential signage.
- Interaction that does not require continuous screen attention.
- Haptic feedback that remains perceptible in noisy environments.

Emotional & Social Needs

- Avoid an overly medical appearance.
- Avoid unnecessarily revealing disability or private information.
- Fit naturally with personal style, clothing

IDEATION

Traditional navigation apps are not enough, so we need to use first-person-view image recognition of the surroundings.

We need wearable devices and enables human-computer interaction via multi-sensory signals beyond vision.

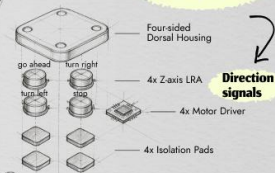
It should function like an everyday accessory — natural to use, portable, and even somewhat fashionable.



Camera clip + voice module

Sunglasses is existing daily eyewear, no extra hardware added. The portable camera clip captures the user's first-person perspective, while the voice module supports real-time voice interaction.

AI processes visual information triggered by voice-input prompts.



This portable strap-form design occupies minimal palm-tactile capacity.

Why reject minimalist ring or fitness-band forms?

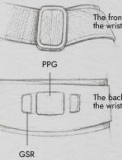
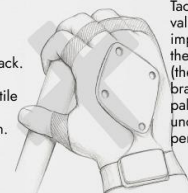
While they minimize medical-device aesthetics, wrists are poor for clear directional vibration feedback. Research shows the back-of-hand has duller tactile perception and prevents excessive haptic stimulation.

Why not a full-enclosure glove structure?

Tactile sensation is extremely valuable for people with visual impairments, we need to leave the fingertip of the index finger (the most sensitive) exposed for braille reading. Meanwhile, the palm should also remain uncovered for object perception and grasping.

Why reject the strap-style cut-out glove structure?

The integrated form offers limited flexibility and cannot fit all users with a single size. In addition, thin straps are prone to damage.



System Design

00 WHY THIS SCENARIO?

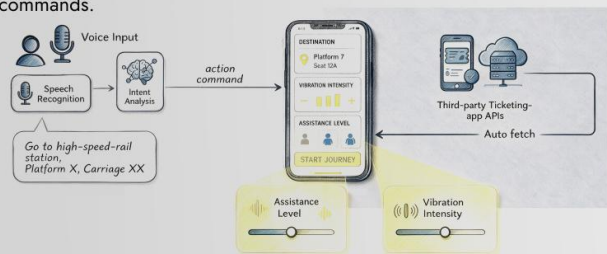
Common outdoor scenarios are supported by Google Maps through external API navigation inputs. **By contrast**, underground parking lots and subway stations depend heavily on way-finding signs and landmarks for directional guidance, creating new obstacles for low-vision users.

This Project target GPS-unavailable environments, solution integrates **first-person vision data + AI + wearable multi-sensory** hardware to enable independent mobility for low-vision people.

01 SET

Before-Trip

- Voice-based destination & requirement setup. Data can also be auto-fetched via third-party ticketing-app APIs.
- Personalisation: adjust assistance level and vibration intensity using voice commands.



This scenario is designed exclusively for low-vision users with residual eyesight (see previous user case).



01 SET



02 SENSE



03 TURN



04 ALERT



05 IDENTIFY

When visual recognition detects obstacles (stairs, sudden pedestrians or objects) and triggers an alert to tell the user to stop.

Voice confirmation will be triggered when the user approaches the destination.



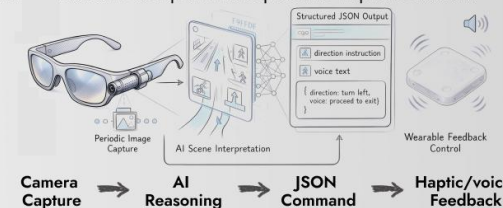
06 ARRIVE

02 SENSE - 03 TURN

During-Trip

Upon entering GPS-denied areas, the camera integrated into the sunglass temple activates and captures images at periodic intervals.

AI analyses the data and outputs structured JSON data with direction instructions and voice text. The wearable device receives signals and drives electronic components to produce haptic feedback.

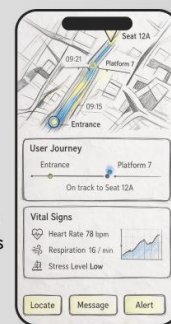


06 ARRIVE

After-Trip

Upon arrival, navigation task ends. Trajectory logs are saved in the backend and shown on mobile for family/caregiver access.

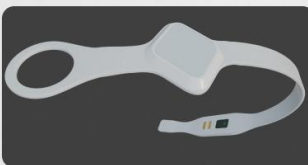
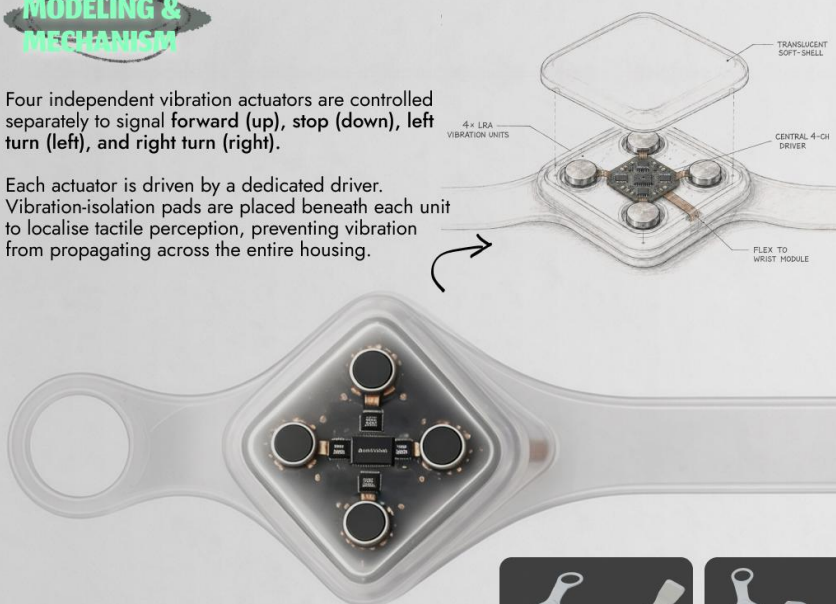
The device also monitors the user's physiological status in real-time. Alerts will be sent to family members if abnormal vital signs are detected.



MODELING & MECHANISM

Four independent vibration actuators are controlled separately to signal forward (up), stop (down), left turn (left), and right turn (right).

Each actuator is driven by a dedicated driver. Vibration-isolation pads are placed beneath each unit to localise tactile perception, preventing vibration from propagating across the entire housing.



EXPERIMENT & CODING

Due to component availability constraints, this experiment was conducted solely on the four-directional haptic vibration module

Current component choices only satisfy basic functionality. Do not fit wearable size

Code testing uses VS Code with the PlatformIO extension, where serial commands simulate large language model inputs.



MCU
ESP32-C3 SuperMini



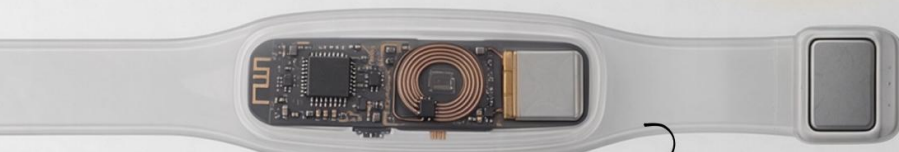
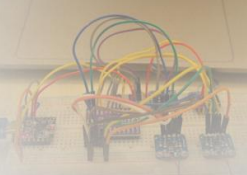
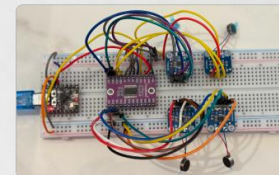
I2C Multiplexer
PCA9548A



Haptic Drivers
DRV2605L X4



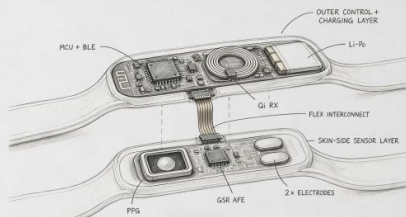
Z-axis Vibration Unit
LRA-0825 X4

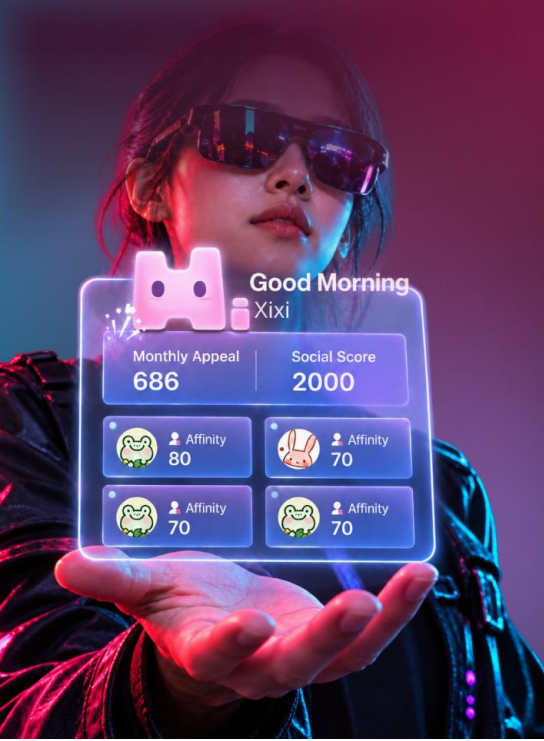


The wrist-unit consists of two layers.

The upper layer houses the main controller and wireless charging module.

The lower layer (skin-facing side) accommodates two physiological sensors (PPG + GSR).





HiMeet: an AR-Assisted Intervention System for People with Social Anxiety

2023 Graduation project



Research

1 DISCOVERY · CAUSAL MODEL

Question

Q1: Why do heavy online socialisers tend to have smaller offline social circles and show more social-anxiety-related behaviours?

Q2: Does social anxiety explain the link between weaker real-world support and greater online social engagement?

User questionnaire statistics



Key Findings

- Weaker real-world social support was associated with higher social anxiety and greater dependence on online socialization.
- Young adults aged 18–25 were especially vulnerable

Design implication: digital support should reconnect people with real-world relationships.

2 PERSONAL SAMPLING · ROLE-PLAY test

Method Self-Report + physiological measures+ video observation

Experimental Procedure Individual session: 45–60 min

1. Read pre-experiment information (2 min) → consent confirmation
2. Social anxiety screening questionnaire (5 min)-LSAS(Liebowitz Social Anxiety Scale)
3. Task & scenario briefing (2 min)
4. Self-perception scale (3 min)
5. Wear EDA wristband for baseline physiological recording (3 min)
6. Situational exposure task (15–20 min)
7. Subjective & somatic self-rating scales (3 min)
8. Semi-structured interview (15 min)



Scenario Tasks

- Enter the room and greet others
- Take a seat
- Introduce yourself appropriately
- Participate in discussion and share opinions
- Initiate one-on-one conversation and exchange contacts

Results

38.0 vs 8.8

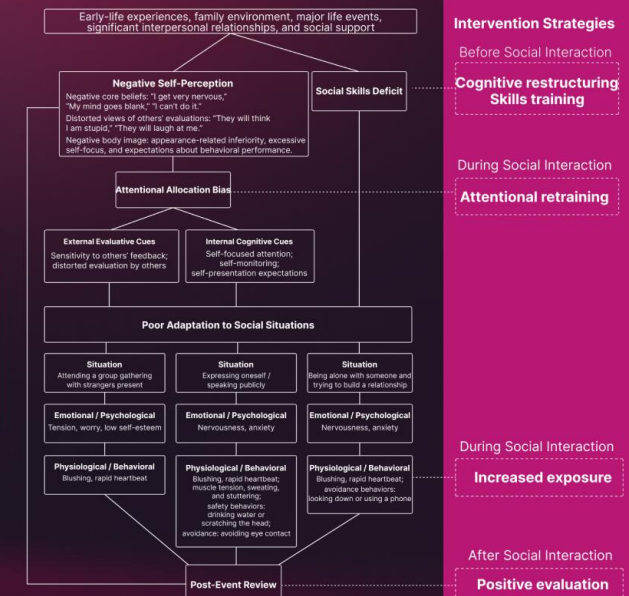
Exposure scenarios is effective: socially anxious participants showed significantly higher subjective distress, $p < .001$

$Y=2.037^*x$

Social anxiety(Y) explained(45.1%) by negative self-cognition(x), especially in negative body-image perceptions, inferiority etc.

Cognitive-behavioral Framework

This test explores cognitions, emotions and behaviours of socially anxious youth in social exposure, and establishes a cognitive-behavioural model, proposing universal intervention strategies targeting frustration triggers.



Design

AR interface design

1 FIRST MEET

The system uses facial recognition to request data from the backend, retrieves personal information, and renders it on the AR device.

Users' social information such as nickname, star sign and occupation is pre-configured and stored in the mobile app.



Try asking about recent TV shows or films he have watched. Be brave and start a new conversation~

2 AFTER CHATTING FOR A WHILE

During social interaction, the camera captures facial expressions in real time to recognise emotions. These emotional cues are fed into the AI as auxiliary input for reasoning.



He looks happy. Your conversation is going well !



DURING

Participate

- Contextual prompts
- Social & emotional cues
- AI for encouraging whisper
- Gesture-based interaction
- Game for encounter

AI+AR

AFTER

Sustain

- Relationship reminders
- Limited messaging online

MOBILE

3 MAKE A CONNECTION

The camera detects human interaction gestures. For instance, it automatically triggers the friend-add action when a handshake is recognised.

Similarly, when the system detects a waving gesture directed toward the user, it displays the prompt: "XX is greeting you; you may try waving back."



Congratulations! You are now friends. Keep in touch~

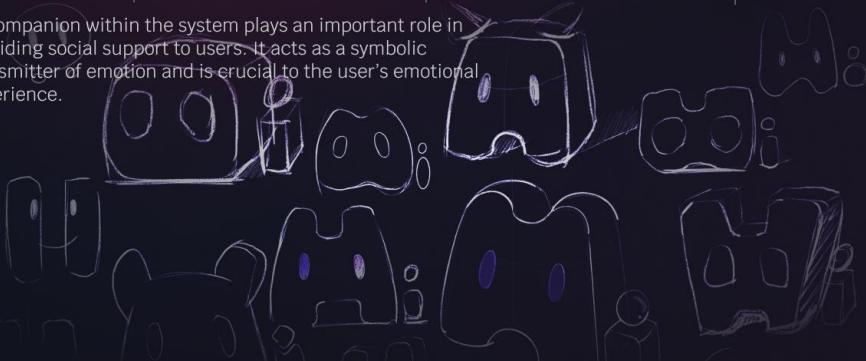


Design

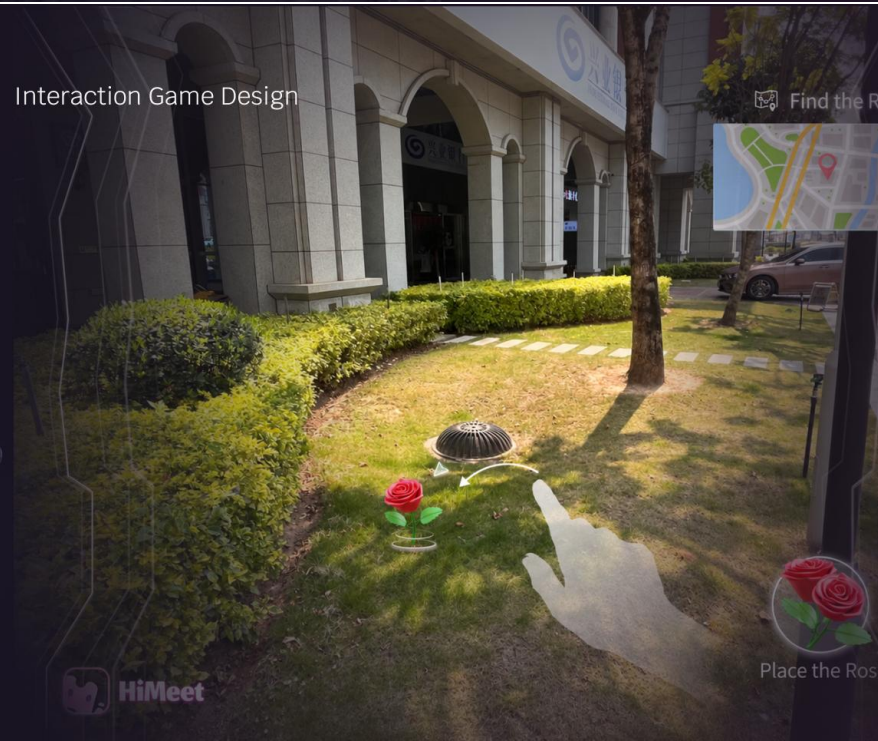
Identity Design



AI companion within the system plays an important role in providing social support to users. It acts as a symbolic transmitter of emotion and is crucial to the user's emotional experience.

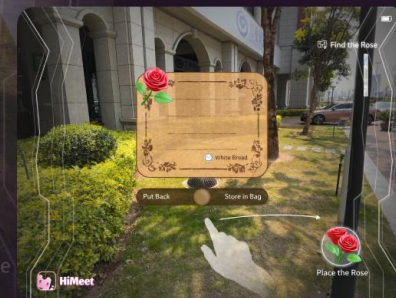


Interaction Game Design



"ROSE ENCOUNTER"

an LBS-powered AR interactive game. Users hunt for roses on a digital map. Nearby roses trigger navigation cursors and distance prompts. Tap 3D virtual roses in physical space to collect them into inventory; drag and release the bottom-right icon in the air to place roses. All operations rely on gestures.



星律

"Worry-relieving Planet"

Loading...

A Playable Game Bracelet

emotion regulation

Wearable Design

Game develop

2020

The 17th "Challenge Cup" Extracurricular Academic and Technological Works Competition

Woven fabric

Detachable sensor module

hook-and-loop Fastener

Motivation

Over 100 million people in China live with mental disorders

1 Insufficient interventions for mild mental health issues

Most hospitals focus on severe mental illnesses, lacking support for people with mild psychological struggles. Many fail to recognise their conditions, and social stigma keeps consultation rates low.

2 Lacking funny Digital mental health tools

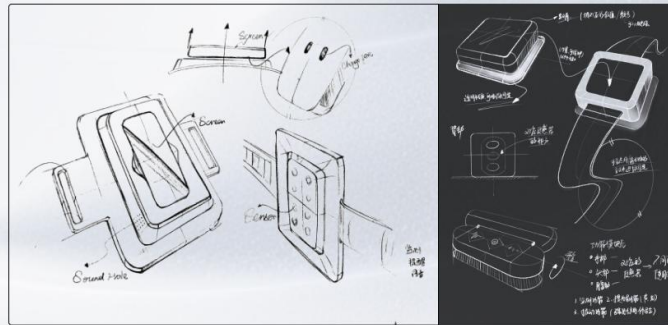
Mental health challenges are affecting more young people. Yet most online tools are serious medical aids, with few playful interactive designs for the general public.

Target

User
living with common mental health concerns including anxiety, sleep disorders and mild depression

Features
combining smart wearables and online platform.
Using interactive games and real-time biometric monitoring, it delivers status tracking, alerts and auxiliary psychological intervention.

Ideation



1 Draft 1

Detachable sensing modules

"Smartwatch + abdominal breathing belt"
The core module of the smartwatch is detachable. It can be mounted onto the abdominal breathing belt to enable respiration-sensing games.

3 Draft 3

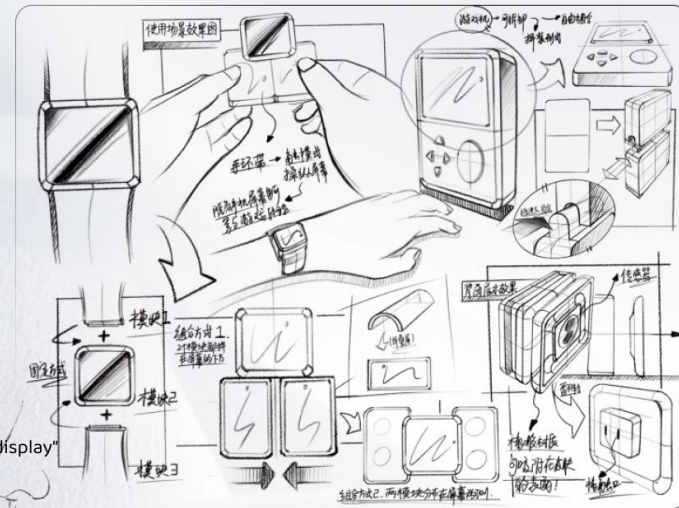
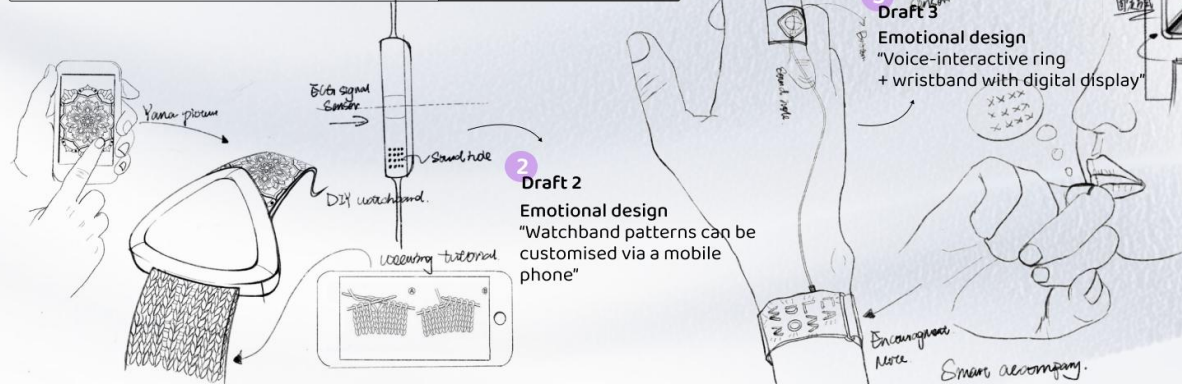
Emotional design "Voice-interactive ring + wristband with digital display"

2 Draft 2

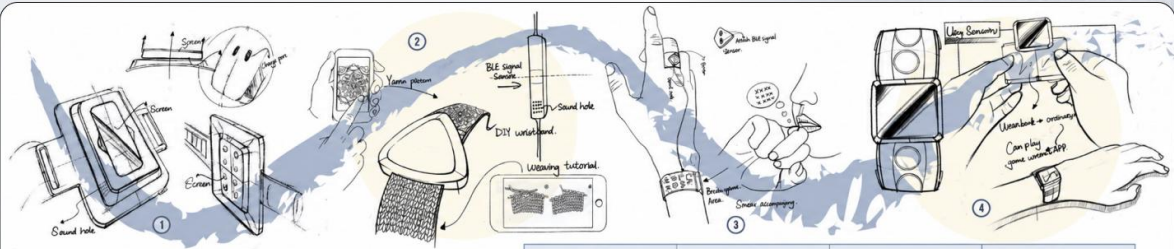
Emotional design "Watchband patterns can be customised via a mobile phone"

4 Draft 4

Gamified design, Inspired by detachable Switch controllers. Each strap acts as an independent digital module. Two straps can be reassembled with the central display module to form a compact game console. It supports standalone gameplay without a mobile phone, enabling gaming anytime and anywhere.



Ideation



Concept selection

Selection criteria	Weight	①	②	③	④
Monitoring accuracy	15%	5	4	4	4
Hardware-app integration	10%	2	4	2	5
Engagement of hardware interaction	15%	2	4	3	5
Technical cost and complexity	10%	2	3	3	3
Additional positive effects (e.g., encouraging positive social interaction)	15%	3	3	4	5
Additional negative effects (e.g., causing embarrassment)	15%	4	4	4	5
Differentiation from existing products	10%	3	1	3	4
Price and difficulty of adoption	10%	4	2	2	4
Total score	100%	3.4	3.4	3.1	4.3
Ranking		2	3	4	1
Continue?		✗	✗	✗	✓

This sheet presents a concept-selection matrix for four design concepts. Eight weighted evaluation criteria covering monitoring accuracy, hardware-app integration, interactive engagement, technical complexity and cost, positive/negative side effects, product differentiation, and adoption barriers. Concept 4 achieved the highest total score of 4.3 and ranked first, so it was selected for further development.

Definition



Game design Games developed in Unity



The game offers three modes: 1-minute trial version, 5-minute basic session, and 10-minute advanced session.

The player's abdominal breathing generates periodic pressure changes inside the pneumatic bladder. The system measures the frequency and intensity of diaphragmatic breathing. Unity reads bladder pressure data through serial communication to control the whale's vertical position.

Players guide the whale by adjusting their breathing rhythm.

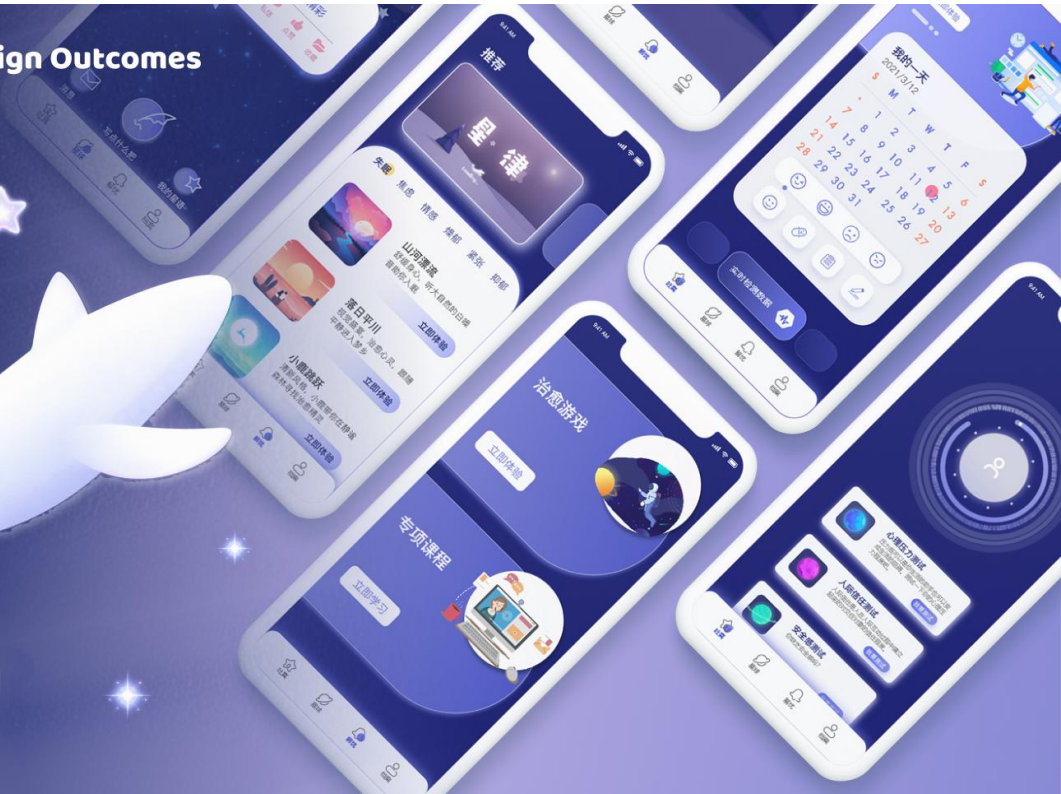


Each wave peak in the game interface contains 16 stars. A full cycle spans 30 seconds. The system counts how many stars the whale collects.

If fewer than 24 stars are gathered, a prompt pops up: "Try adjusting your breathing to keep up with the rhythm", guiding players to correct their breathing in real time.



APP Design Outcomes



Find It



Campus Pickup Service System Design

Enterprise project

2021

National 3rd Prize in UXDA (International User Experience Innovation Competition)
Part of the system solution was adopted by Alibaba's courier service app team for experimental grayscale testing.



THE PROBLEM: CAMPUS PICKUP

People aged 19-30 make up 37% of online shoppers; students and young workers are key buyers. Campus pickup has distinct needs compared with other communities.

Many users
Many people and parcels
Sales events bring a surge

Busy pickup times
Similar pickup times
lead to congestion

Restricted access
Students must collect
their own parcels

KEY PROBLEMS TO SOLVE

Most couriers are moving toward smarter last-mile services. Cainiao serves 57% of campuses, yet satisfaction with pickup remains low. User research identified two main problems:

Slow to find
Parcels are hard to locate quickly
Poor shelf signs / crowded parcels / cluttered labels

Slow to check out
Checkout is complicated
Separate ID scans / queues at peak times

SOLUTION

RFID-powered
Efficiency at every step

01 A new campus pickup experience

Smart services at pickup stations
RFID packaging (green flexible printed electronics), parcel tracking, smart recycling bins, and creative reuse displays and exchanges

Online mini app
3D shelf and parcel previews, AR navigation, recycling rewards, social play

02 Parcel information in the cloud

Smarter shipping labels and protect personal information. Manage data by delivery stage (province, city, local area and last mile).

03 Recycling and social features

Use the campus community and points to reward recycling and reuse. Offer a place to exchange items, with fun features that add value.

DESIGN VALUE

A better pickup experience
Recycling awareness and social connection

CAMPUS PARCEL PICKUP TODAY

(Desk research, field visits and the first survey)

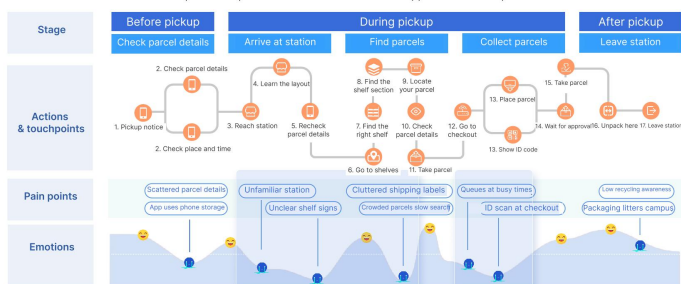
We studied campus pickup habits, problems and needs through field visits and a survey (24 universities; 270 valid responses).
Most couriers favor smart last-mile services. Cainiao serves 57% of campuses, but user satisfaction remains low.



USER RESEARCH

User Journey Map

Map each step: moments of frustration reveal opportunities to improve



USER RESEARCH

(Second survey and in-depth interviews)

Survey

A SWTH survey focused on pickup at Cainiao stations

Who: Gender, school, occupation, year, age, shopping frequency, box disposal
What: Satisfaction with pickup at Cainiao stations
Where: Pickup locations
When: Pickup times
Why: Reasons for low satisfaction
How: Suggestions for better campus pickup

64.6%



Liu Duoduo Female / 20 / Student
Frequent online shopper
Collects parcels in batches
Picks up for classmates
Tracks spending

Pain points

1. Many parcel types make searching difficult.
2. Each parcel needs a separate checkout.

User needs

1. Know exactly where each parcel is.
2. Collect several parcels faster.
3. Understand personal shopping habits.

In-depth Interviews

8 people with different profiles, all unhappy with Cainiao pickup
Explore pickup habits, feelings, good and bad experiences, and suggestions.

35.4%



Zhang Mangmang Male / 24 / Grad student
Shops for essentials
Eco-conscious
Busy; dislikes queues

Pain points

1. Late classes often mean missing opening hours.
2. Long queues during busy periods.

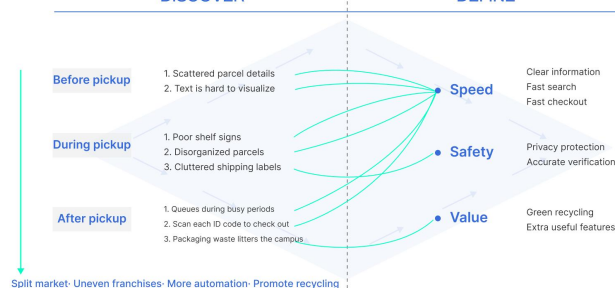
User needs

1. 24-hour pickup.
2. Faster pickup at peak times.
3. Safe, convenient pickup help.

KEY USER NEEDS

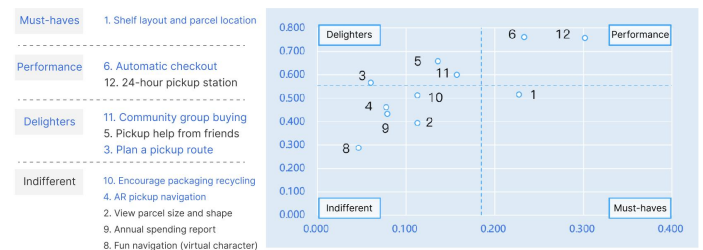
DISCOVER

DEFINE

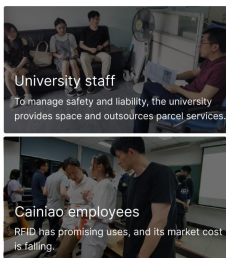


KANO NEEDS ANALYSIS

● Planned ● Removed / On hold

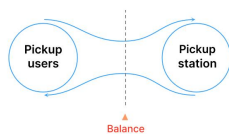


STAKEHOLDER INTERVIEWS



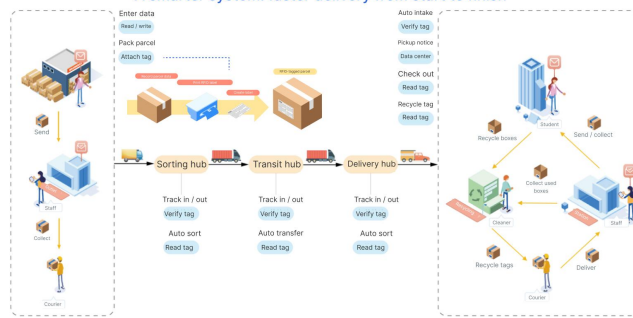
We interviewed university staff and Cainiao employees to explore solutions. For station managers, cost comes first. They often meet only basic pickup needs. As the industry becomes smarter, change is essential and user experience is receiving more attention.

When profit and user experience conflict, we need to balance station profitability with a better pickup experience.



SMART PARCEL PICKUP WITH RFID

A smarter system: faster delivery from start to finish



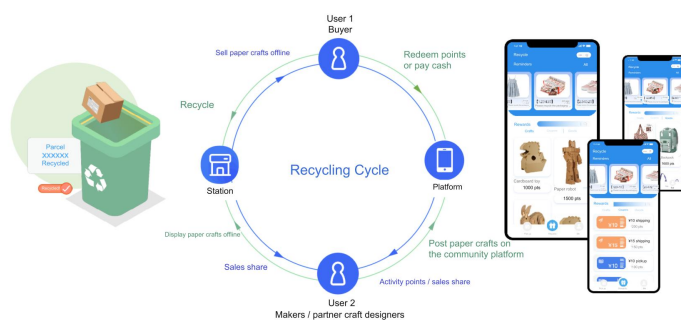
ENGAGEMENT STRATEGY

Use the campus community to encourage recycling and reuse



Recycling + Social Play

RECYCLING CYCLE



SOCIAL FEATURES

V-characters Interface

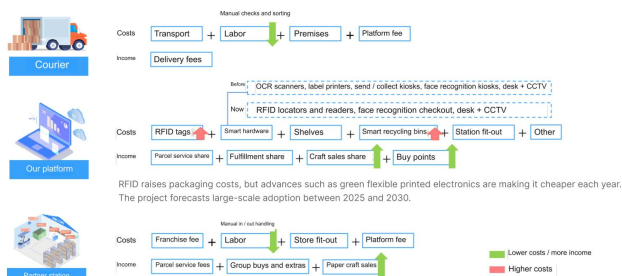
Shopping habits shape your profile and avatar. Use points to dress your avatar and raise your style score. Campus leaderboards let you compete with other students. Shake your phone to find nearby players for a style challenge. Winners have a chance to earn bonus rewards.

Social connection · Fun interaction · More spending · Repeat visits



COST AND REVENUE ANALYSIS

Lower labor costs, new ways to trade recycled goods, and revenue from points purchases



DELIVER

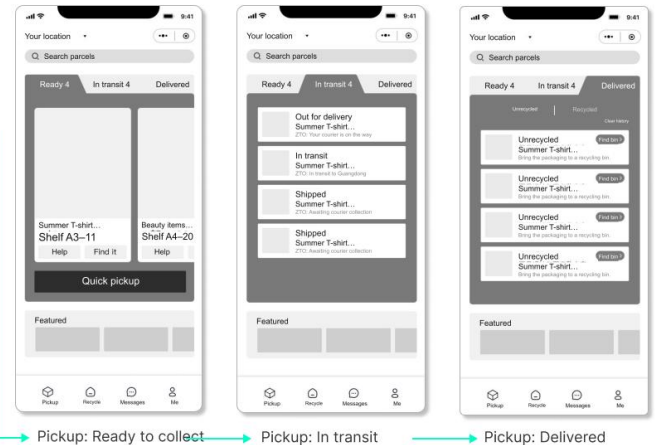
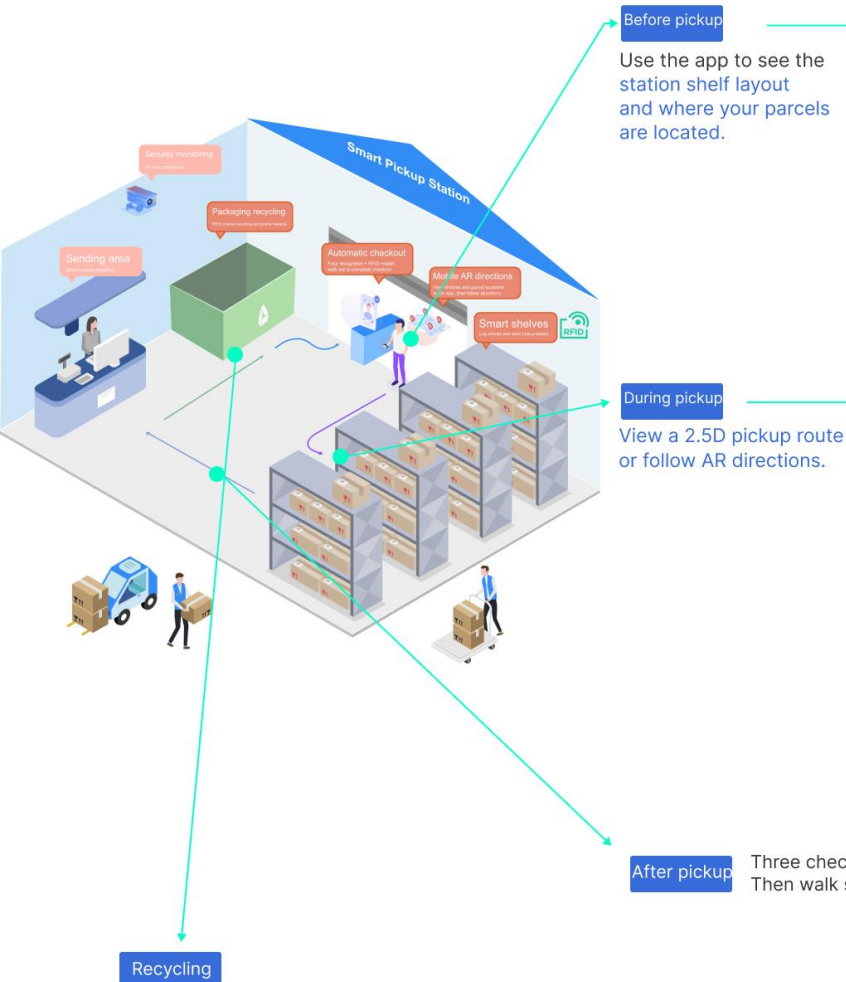
Smart Parcel Pickup with RFID

Smart station + Mini app = A better last-mile pickup experience

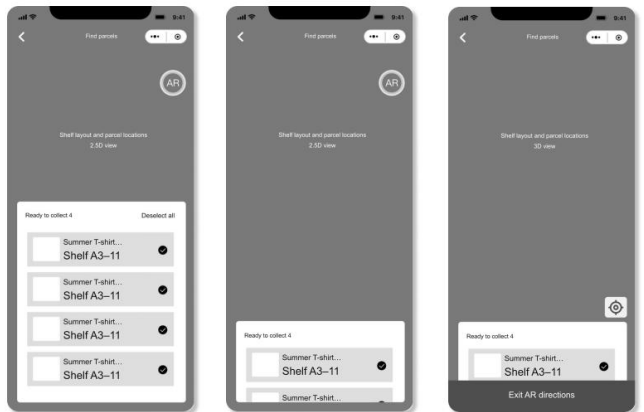
Pickup Station

User Actions

Key Interactions



Pickup: Ready to collect → Pickup: In transit → Pickup: Delivered



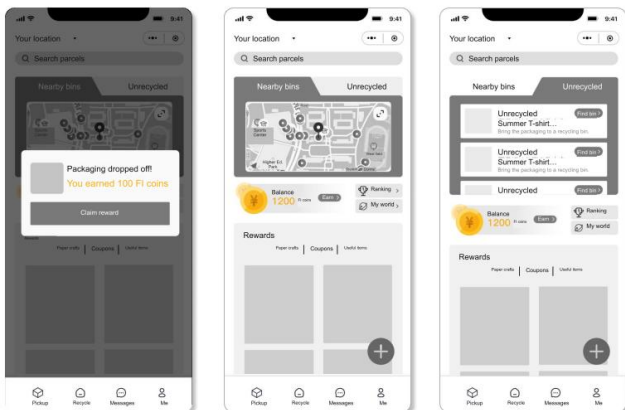
After pickup Three checks for accuracy: When you take a parcel, verify it is yours in the app. Then walk straight out of the station to finish checkout.

App: RFID + phone sensors verify ownership — self-check
Station: RFID reader + face recognition at the exit — theft prevention
Data: The shipping label includes a nickname and the last 4 phone digits

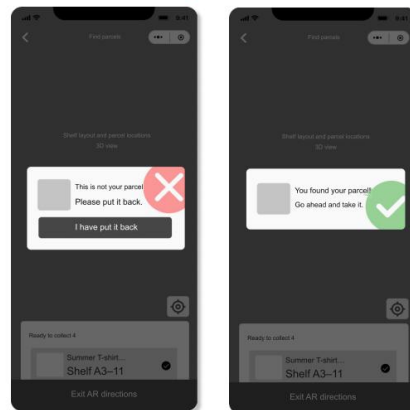
Recycling

Place RFID-tagged packaging in a recycling bin and earn rewards.

Recycling reward pop-up ← Nearby bins ← Unrecycled parcels



Wrong parcel alert ← Correct parcel toast





AUTOMOTIVE HMI DESIGN

Dashboard & Central Control Home Screen

2022 UI Practice

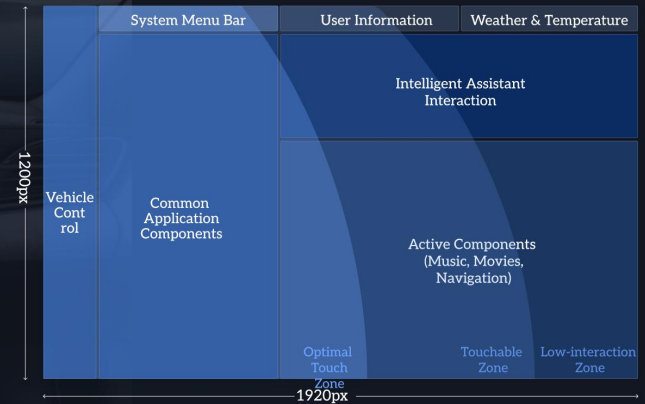
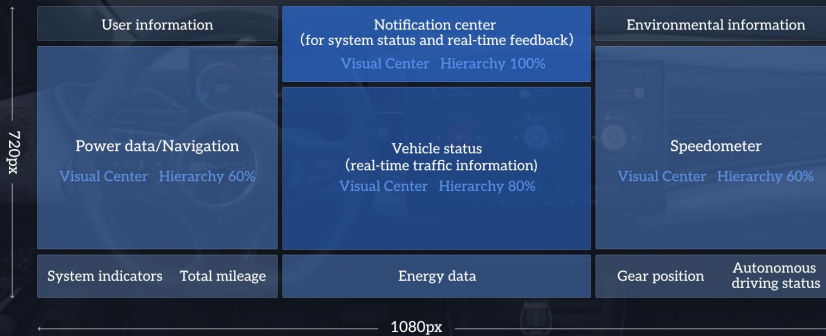
This design follows ergonomics, cognitive psychology and safe-driving principles to deliver a dynamic interface system with a minimalist aest



Dashboard

Framework

Functional zoning and information hierarchy



DESIGN OUTCOMES

High-fidelity mockups for selected pages

Notification center

Serves as a key area for **real-time status and information feedback**.

AUI

The interface **adaptively** displays functions based on real-time conditions

